

E - Crossing Table Cloth

分值: 475 分

Problem Statement

There are N cells arranged in a horizontal row. The i -th cell from the left ($1 \leq i \leq N$) is called cell i .

水平排列着 N 个格子。从左数的第 i 个格子 ($1 \leq i \leq N$) 称为格子 i 。

There are M pieces of cloth. Laying cloth i ($1 \leq i \leq M$) covers cells L_i through R_i .

共有 M 块布。铺设布 i ($1 \leq i \leq M$) 会覆盖从 L_i 到 R_i 的所有格子。

Answer Q queries. For the q -th query ($1 \leq q \leq Q$), integers S_q and T_q are given, so answer the following problem.

你需要回答 Q 个询问。对于第 q 个询问 ($1 \leq q \leq Q$)，会给出整数 S_q 和 T_q ，请回答如下问题。

- Determine whether it is possible to choose exactly two pieces of cloth from the M pieces and lay them so that the following condition is satisfied.
 - Cells S_q through T_q are covered by at least one piece of cloth, and no other cells are covered by any cloth.
- 判断是否可以从 M 块布中恰好选择两块并铺设，使得满足以下条件：
 - 从 S_q 到 T_q 的所有格子都被至少一块布覆盖，且除此之外没有其他格子被任何布覆盖。

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $2 \leq M \leq 2 \times 10^5$
- $1 \leq L_i \leq R_i \leq N$
- $1 \leq Q \leq 2 \times 10^5$
- $1 \leq S_q \leq T_q \leq N$
- All input values are integers.

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- $2 \leq M \leq 2 \times 10^5$
- $1 \leq L_i \leq R_i \leq N$
- $1 \leq Q \leq 2 \times 10^5$
- $1 \leq S_q \leq T_q \leq N$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
N M
L1 R1
L2 R2
⋮
LM RM
Q
S1 T1
S2 T2
⋮
SQ TQ
```

Output

Output the answers for the queries, separated by newlines.

输出每个询问的答案，用换行分隔。

For each query, output Yes if it is possible to choose two pieces of cloth satisfying the condition, and No otherwise.

对于每个询问，如果存在满足条件的两块布则输出 Yes，否则输出 No。

Sample Input 1

```
4 3
1 3
1 1
2 4
4
1 4
2 4
1 3
1 1
```

Sample Output 1

```
Yes
No
Yes
No
```

For the first query, the condition can be satisfied by choosing cloth 1 and cloth 3.

对于第一个询问，选择布 1 和布 3 即可满足条件。

For the third query, the condition can be satisfied by choosing cloth 1 and cloth 2.

对于第三个询问，选择布 1 和布 2 即可满足条件。

For the second and fourth queries, no choice of two pieces of cloth can satisfy the condition.

对于第二个和第四个询问，不存在任何选择两块布的方案能满足条件。

Sample Input 2

```
7 10
2 6
2 5
3 6
1 6
1 2
5 6
```

```
2 3
3 7
2 3
1 2
10
1 2
3 5
1 4
1 5
1 5
5 7
1 6
2 3
5 7
2 4
```

Sample Output 2

```
Yes
No
No
Yes
Yes
No
Yes
Yes
No
No
```